



Fourier Integral Representation

Analysis of a Truncated Cosine Wave

Abstract

This report details the mathematical formulation and numerical evaluation of the Fourier integral for a non-periodic, piecewise continuous signal. The continuous frequency spectrum is derived, and the original signal is numerically reconstructed to demonstrate the efficacy of the Fourier transform approach.

1 Mathematical Formulation

Unlike periodic signals that are represented by discrete Fourier series, non-periodic signals defined over $(-\infty, \infty)$ are represented by the continuous Fourier integral.

Consider the piecewise function $f(x)$ defined as:

$$f(x) = \begin{cases} \cos(x) & 0 < x < \pi \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

The Fourier integral representation of $f(x)$ is given by:

$$f(x) = \int_0^\infty [A(\omega) \cos(\omega x) + B(\omega) \sin(\omega x)] d\omega \quad (2)$$

where the spectral coefficient functions $A(\omega)$ and $B(\omega)$ are defined via integration over the domain where the signal is non-zero:

$$A(\omega) = \frac{1}{\pi} \int_0^\pi \cos(x) \cos(\omega x) dx \quad (3)$$

$$B(\omega) = \frac{1}{\pi} \int_0^\pi \cos(x) \sin(\omega x) dx \quad (4)$$

1.1 Evaluation of Spectral Coefficients

Using trigonometric product-to-sum identities, we evaluate $A(\omega)$. For $\omega \neq 1$:

$$A(\omega) = \frac{1}{2\pi} \int_0^\pi [\cos((1-\omega)x) + \cos((1+\omega)x)] dx \quad (5)$$

Evaluating the integral at the boundaries 0 and π :

$$A(\omega) = \frac{1}{2\pi} \left[\frac{\sin((1-\omega)\pi)}{1-\omega} + \frac{\sin((1+\omega)\pi)}{1+\omega} \right] \quad (6)$$

Using the properties of the sine function ($\sin(\pi - \omega\pi) = \sin(\omega\pi)$ and $\sin(\pi + \omega\pi) = -\sin(\omega\pi)$), this simplifies to:

$$A(\omega) = -\frac{\omega \sin(\omega\pi)}{\pi(\omega^2 - 1)} \quad (7)$$



Similarly, for $B(\omega)$ when $\omega \neq 1$:

$$B(\omega) = \frac{1}{2\pi} \int_0^\pi [\sin((1+\omega)x) - \sin((1-\omega)x)] dx \quad (8)$$

$$B(\omega) = \frac{\omega \cos(\omega\pi) + 1}{\pi(\omega^2 - 1)} \quad (9)$$

2 Computational Implementation

To numerically validate the theoretical derivation, a MATLAB script was developed to compute the continuous spectrum and reconstruct the signal. The inverse Fourier integral is approximated by integrating over the finite frequency range $\omega \in [0, 25]$, as specified by the problem scope.

```

1 % Reconstruct the Signal using the Fourier Integral
2 for k = 1:length(x_plot)
3     x_val = x_plot(k);
4     % Define integrand: A(w)cos(wx) + B(w)sin(wx)
5     reconstruction_integrand = A_w .* cos(w * x_val) + B_w .* sin(w * x_val)
6     );
7     % Integrate over frequency domain using trapezoidal rule
8     f_reconstructed(k) = trapz(w, reconstruction_integrand);
9 end

```

Listing 1: Numerical Signal Reconstruction via Fourier Integral

The resulting visual analysis confirms that integrating up to $\omega = 25$ yields a highly accurate reconstruction of the original truncated cosine pulse, with minor high-frequency ripple artifacts near the discontinuities at $x = 0$ and $x = \pi$.